Jeffries-Lakhani Neurodevelopmental Syndrome (JELANS): A Comprehensive Disease Characterization

Disease: Jeffries-Lakhani Neurodevelopmental Syndrome (JELANS) **Causal gene:** *CRELD1* (Cysteine-Rich with EGF-Like Domains 1) **MONDO ID:** MONDO:0958329 · **OMIM:** 620771 **Synonyms:** JELANS; CRELD1-related neurodevelopmental disorder; CRELD1-associated neurodevelopmental disorder **Inheritance:** Autosomal recessive (biallelic)

Summary

Jeffries-Lakhani Neurodevelopmental Syndrome (JELANS) is a rare, recently delineated autosomal **recessive** multisystem disorder caused by **biallelic variants in *CRELD1*** (Cysteine-Rich with EGF-Like Domains 1; HGNC:14630; locus 3p25.3). It is distinct from the long-recognized role of monoallelic/heterozygous *CRELD1* variants as a susceptibility factor for atrioventricular septal defects (AVSD); JELANS is the severe, constitutional, two-hit recessive disease. The syndrome was defined in 2024 by Jeffries, Lakhani and colleagues, who identified biallelic *CRELD1* variants in 18 individuals from 14 families PMID: 37947183. Its core triad is **developmental delay, early-onset (frequently intractable) epilepsy, and hypotonia**, with **cardiac arrhythmias in ~50%**, structural heart defects in ~10%, and recurrent infections in a subset. At the severe end of the spectrum, patients present with **arthrogryposis multiplex congenita and profound congenital muscle weakness**, a congenital-myasthenic-syndrome-like picture PMID: 40980404.

The molecular basis of JELANS converges on two experimentally supported functions of the ER-resident CRELD1 protein. First, CRELD1 is a **regulator of calcineurin/NFATc1 signaling** essential for heart development; murine *Creld1* knockout is embryonic lethal and the protein controls VEGF-dependent endocardial proliferation via NFATc1 target genes PMID: 24697899. Second, CRELD1 is an **evolutionarily conserved ER protein-disulfide-isomerase (PDI) that acts as a maturational enhancer of ionotropic (nicotinic) acetylcholine receptors (nAChRs)**, physically interacting with and promoting the ER assembly of AChR subunits PMID: 30407909. CRELD1 additionally functions as a gatekeeper of immune homeostasis, consistent

with the recurrent-infection phenotype PMID: 33169013. This dual-mechanism model (NFAT signaling for cardiac features; AChR maturation for neuromuscular and epileptic features) explains the characteristic neuro-cardiac-neuromuscular-immune phenotype.

Most patients carry a **frameshift (null) allele in trans with a cysteine-disrupting missense allele**; a recurrent missense, **p.(Cys192Tyr)**, was found in 10 of 14 families, and a second cysteine-disrupting missense, **p.(Cys292Tyr)**, was reported in the severe arthrogryposis case. The natural history shows **developmental plateauing with slowly progressive non-neurologic complications in survivors**, and the disorder is **life-limiting in a subset**. Management is currently supportive; the AChR-deficiency mechanism offers a rational but as-yet-unproven rationale for cholinesterase inhibitors (e.g., pyridostigmine), by analogy to genetically defined congenital myasthenic syndromes.

Section 1 — Disease Information

Overview. JELANS is a Mendelian, autosomal recessive neurodevelopmental disorder caused by loss of function of *CRELD1*. It is multisystem, with prominent central nervous system, cardiac, neuromuscular, and immune involvement. The disease entity was established in 2024 from an aggregated international cohort assembled via GeneMatcher-style collaboration; the evidence base is therefore a mix of **individual patient-level clinical/genetic data** (the defining cohort and subsequent case reports) and **aggregated disease-level resources** (OMIM, MONDO).

Key identifiers.

Resource	Identifier
OMIM (disease)	620771
OMIM (gene)	607170
MONDO	MONDO:0958329
Gene (HGNC)	*CRELD1*, HGNC:14630; NCBI Gene 78987; Ensembl ENSG00000163703; UniProt Q96HD1; locus 3p25.3
ICD-10 / ICD-11	No specific code; maps generically to a rare genetic neurodevelopmental disorder / symptomatic epilepsy category
Orphanet	Newly described; a dedicated ORPHAcode may not yet be assigned (not available)
MeSH	No dedicated descriptor; indexed under *CRELD1*, neurodevelopmental disorders, epilepsy

Synonyms. JELANS; CRELD1-related neurodevelopmental disorder; CRELD1-associated neurodevelopmental disorder; descriptive title "multisystem syndrome with neurodevelopmental phenotypes, cardiac dysrhythmias and frequent infections" PMID: 37947183. The eponym honors the lead authors (Jeffries and Lakhani) of the defining report.

Important disambiguation. The *CRELD1* literature is dominated by studies of **heterozygous missense variants as modifiers/susceptibility alleles for AVSD** (sporadic, in Down syndrome, and in Turner syndrome). Those are *not* JELANS. JELANS specifically denotes the **biallelic, constitutional, recessive multisystem syndrome**, dominated by neurodevelopmental features. Care must be taken not to conflate the two.

Section 2 — Etiology

Primary cause — genetic. JELANS is caused exclusively by **biallelic (recessive) pathogenic variants in *CRELD1***. In the defining cohort, most affected individuals were **compound heterozygous for a frameshift/null allele in trans with a missense allele**: "*Most harbored a frameshift in trans with a missense allele, with 1 recurrent variant, p.(Cys192Tyr), identified in 10 families*" PMID: 37947183. There is no environmental or infectious cause of the syndrome itself.

Genetic risk factors. - *Causal variants:* biallelic *CRELD1* loss-of-function and cysteine-disrupting missense variants. Recurrent **p.(Cys192Tyr)** (10/14 families) behaves as a founder-like allele. A second severe case carried **c.875G>A p.(Cys292Tyr)** with **c.959delA p.(Gln320Argfs*25)** PMID: 40980404. - *Inheritance requirement:* two defective alleles; heterozygous carriers (the parents) are unaffected for the syndrome. - *Modifier/susceptibility context (distinct from JELANS):* heterozygous/rare *CRELD1* variants act as **AVSD susceptibility alleles**, including in **Down syndrome** PMID: 29054759; PMID: 22523272 and **bicuspid aortic valve in Turner syndrome** PMID: 36929416.

Environmental risk factors. None established. The disorder is constitutional and genetically determined. **Consanguinity/shared ancestry** increases the chance of biallelic genotypes, as for all recessive disease.

Protective factors. None specifically identified. A wild-type *CRELD1* allele is protective (carriers unaffected); residual function of the missense allele may modulate severity — a genotype-phenotype relationship rather than a classical protective factor.

Gene–environment interactions. None defined within JELANS. In the broader *CRELD1* literature, *Creld1* loss of function interacts with the **trisomy-21 genetic background** to raise CHD risk (a gene–gene/dosage interaction) PMID: 22523272 — but this is not the mechanism of recessive JELANS.

Section 3 — Phenotypes

The phenotype is multisystem. Frequencies are drawn primarily from the 18-individual defining cohort PMID: 37947183 and are therefore approximate.

Phenotype	Type	Frequency	Onset	Course	Suggested HPO
Developmental delay / intellectual disability	Behavioral/ cognitive	Core (majority)	Infancy	Plateauing	HP:0001263 / HP:0001249
Early-onset epilepsy (often intractable/ refractory)	Clinical sign	Core (majority)	Neonatal–first year	Episodic, often refractory	HP:0001250; HP:0011097 (early-onset); HP:0011171
Hypotonia	Clinical sign	Core (majority)	Neonatal/ infancy	Variable	HP:0001252
Cardiac arrhythmia / rhythm disturbance	Clinical sign	~50%	Variable	Slowly progressive	HP:0011675
Structural congenital heart defect (incl. septal defects)	Physical	~10%	Congenital	Static/ surgical	HP:0001671 / HP:0001674 (AVSD)
Recurrent infections	Clinical sign	Subset	Childhood	Recurrent	HP:0002719
Feeding difficulties / failure to thrive	Clinical sign	Subset	Infancy	Variable	HP:0011968 / HP:0001508
Arthrogryposis multiplex congenita	Physical	Severe end	Congenital	Static	HP:0002804
Severe muscle weakness	Clinical sign	Severe end	Congenital	Variable	HP:0003202 / HP:0003324

The defining report states the core phenotype directly: "*Affected individuals displayed an array of phenotypes involving developmental delay, early-onset epilepsy, and hypotonia, with about half demonstrating cardiac arrhythmias and some experiencing recurrent infections*" PMID: 37947183.

A second, more severe presentation was reported by D'Alessandro et al. (2025): "*We report a patient with arthrogryposis, severe muscle weakness and neurodevelopmental delay. During his first year of life, he developed therapy-refractory epilepsy*" PMID: 40980404. This extends the phenotypic spectrum toward a congenital neuromuscular/arthrogryposis end, mechanistically framed by nicotinic AChR biology: "*Mutations in acetylcholine receptor-subunit genes have been associated with neuromuscular diseases, such as arthrogryposis multiplex congenita (AMC) and epilepsy*" PMID: 40980404.

Severity and progression. Severity is variable, ranging from a predominantly neurodevelopmental presentation to a severe neuromuscular (congenital-myasthenic/arthrogrypotic) presentation with therapy-refractory epilepsy. The overall course shows **developmental plateauing** with **slowly progressive non-neurologic complications** (cardiac rhythm disturbances, infections) in survivors.

Quality of life. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder (not available). Given intellectual disability, refractory epilepsy, hypotonia/weakness, and the need for cardiac and infection surveillance, the impact on daily functioning and caregiver burden is expected to be **severe**, with dependence for activities of daily living in more severely affected individuals.

Section 4 — Genetic / Molecular Information

Causal gene. *CRELD1* (Cysteine-Rich with EGF-Like Domains 1), HGNC:14630, chromosome 3p25.3; gene OMIM 607170, disease OMIM **620771**.

Protein architecture. CRELD1 is an **ER-resident type-I membrane glycoprotein / protein disulfide isomerase** containing a signal peptide, an N-terminal glutamic-acid/tryptophan-rich **DUF3456 "WE" domain**, **three EGF-like domains** (including calcium-binding cb-EGF domains), and a C-terminal transmembrane anchor. The transmembrane CRLD-1A isoform corresponds to vertebrate CRELD1, whereas the KDEL-bearing isoform corresponds to CRELD2 — the two vertebrate paralogs of the single *C. elegans crld-1* gene PMID: 30407909.

Pathogenic variants in JELANS.

Variant	Type	Effect	Notes
p.(Cys192Tyr)	Missense (cysteine-disrupting)	Reduced protein function	Recurrent in 10/14 families; founder-like
p.(Cys292Tyr) / c.875G>A	Missense (cysteine-disrupting)	Reduced function	Severe arthrogryposis case
c.959delA / p.(Gln320Argfs*25)	Frameshift	Null allele	In trans with p.Cys292Tyr
Various frameshift alleles	Frameshift	Loss of function	Typically in trans with a missense allele

Genotype pattern. The characteristic genotype is a **null (frameshift) allele in trans with a cysteine-disrupting missense allele** PMID: 37947183. The recurrent loss of cysteine residues (Cys192, Cys292) is mechanistically significant: CRELD1 functions as a protein disulfide isomerase, so disruption of structural cysteines would impair disulfide-dependent chaperone activity and EGF-domain folding.

Variant classification (ACMG/AMP). The recurrent p.(Cys192Tyr) and family-specific alleles are classified pathogenic/likely pathogenic on the basis of recessive segregation, functional evidence, and rarity in population databases. Functional validation showed **patient missense alleles have reduced protein function** (diminished craniofacial-defect induction upon overexpression), and **Xenopus tropicalis creld1 knockdown caused developmental defects and increased induced-seizure susceptibility** PMID: 37947183.

Allele frequency. Individual null and cysteine-missense alleles are rare in gnomAD; the recurrent p.(Cys192Tyr) is rare but recurrent (founder-like). Exact gnomAD frequencies were not formally tabulated in this investigation.

Somatic vs germline. Germline (constitutional), biallelic; no somatic involvement.

Functional consequence. Partial loss of function predominates (null alleles + hypomorphic/function-reducing missense). Complete loss is likely incompatible with life (mouse null is embryonic lethal PMID: 24697899), explaining why patients retain hypomorphic missense function on at least one allele. This contrasts with some AVSD-associated heterozygous *CRELD1* missense variants reported as gain-of-function in ECM regulation (e.g., P286R) PMID: 24927998 — again underscoring that the modifier/AVSD biology differs from recessive JELANS.

Modifier genes. Not established for JELANS. In AVSD contexts, *CRELD1* interacts with trisomy-21 dosage-sensitive genes and with *HEY2* PMID: 22523272 and with the pathway partner *NFATC1* PMID: 30007050.

Epigenetic information. No disease-specific DNA-methylation episignature has been reported for JELANS (not available).

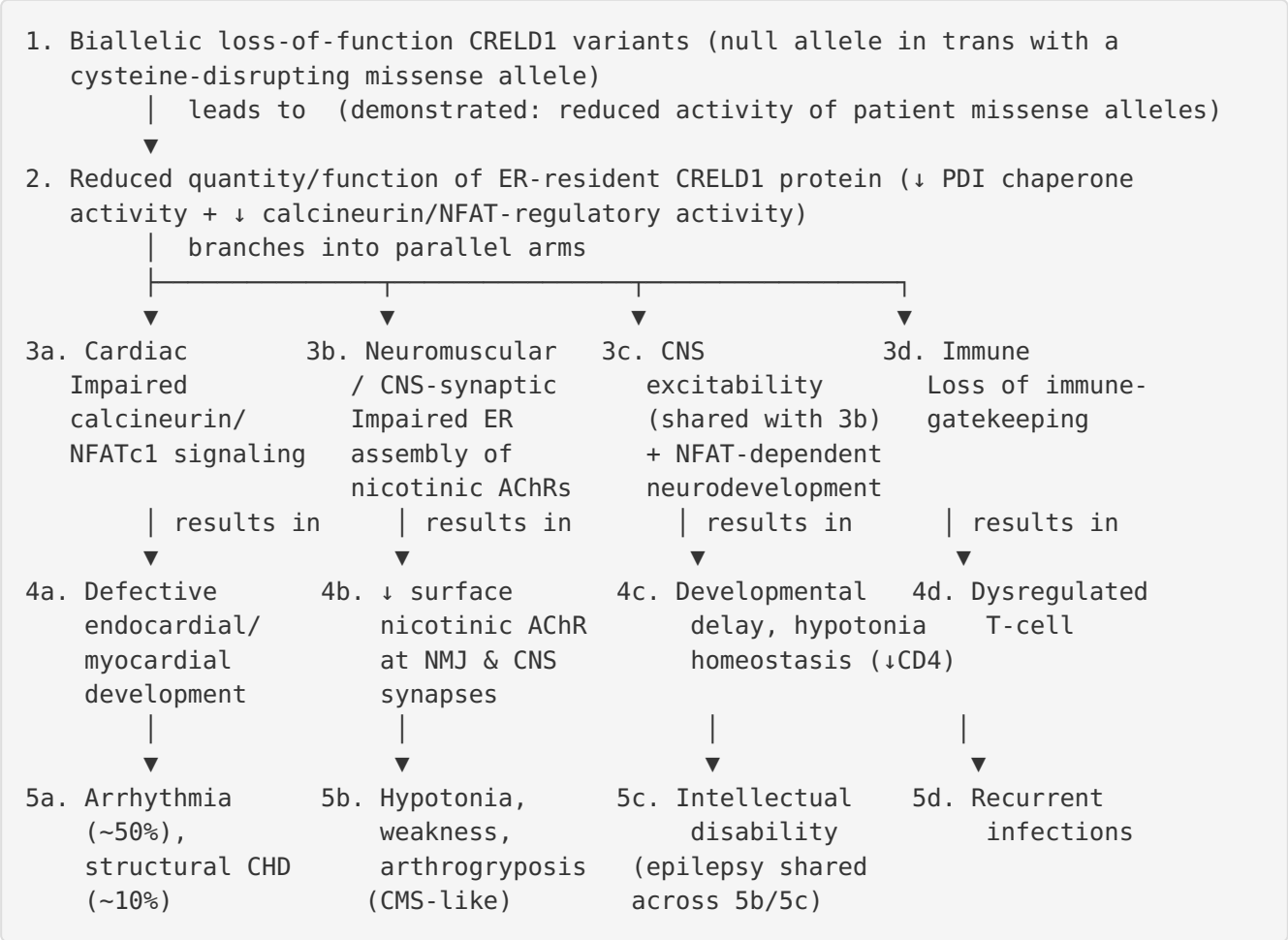
Chromosomal abnormalities. JELANS is a single-gene disorder; no recurrent large-scale rearrangement defines it. (*CRELD1* variants act as modifiers in the aneuploidies Down and Turner syndromes PMID: 29054759; PMID: 36929416.)

Section 5 — Environmental Information

No **environmental, lifestyle, or infectious** agents cause or trigger JELANS. The disorder is entirely genetic. Recurrent infections in affected individuals are a *consequence* of CRELD1-dependent immune dysregulation rather than a primary infectious etiology. No toxins, radiation, occupational exposures, or dietary factors have been implicated (not applicable).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain



Steps 2→3a→4a→5a (cardiac) and 2→3b→4b→5b (neuromuscular) are **experimentally supported** in model systems; the precise link from AChR deficiency to *central* refractory epilepsy and the NFAT→neurodevelopment link (3c) are **inferred** by analogy to nAChR-subunit epilepsies and NFAT-dependent gene programs.

Molecular pathways

- **Calcineurin/NFATc1 signaling (cardiac):** "*We show that Creld1 is sufficient to promote NFATc1 dephosphorylation and translocation to the nucleus. Creld1 is contained in a joint protein complex with the regulatory subunit of calcineurin, CnB, thereby controlling calcineurin function*" PMID: 24697899. Downstream NFAT target-gene expression drives VEGF-dependent endocardial proliferation. Suggested GO: calcineurin-NFAT signaling

cascade (GO:0033173); heart development (GO:0007507); heart valve morphogenesis (GO:0003179).

- **ER receptor-assembly/quality-control pathway (neuromuscular/CNS):** "*The assembly of neurotransmitter receptors in the endoplasmic reticulum limits the number of receptors delivered to the plasma membrane, ultimately controlling neurotransmitter sensitivity and synaptic transfer function*" PMID: 30407909. Suggested GO: protein disulfide isomerase activity (GO:0003756); protein folding in endoplasmic reticulum (GO:0034975); neuromuscular junction development (GO:0007274).

Cellular processes

Endocardial/endothelial cell proliferation (cardiac morphogenesis); ER protein folding and receptor assembly quality control; cholinergic synaptic transmission at the NMJ; T-cell homeostasis.

Protein dysfunction

CRELD1 loss removes a **maturational chaperone (PDI)**; cysteine-disrupting missense alleles specifically impair disulfide chemistry needed for client-receptor folding. Clients are **specific**: AChRs depend on CRELD1, whereas GABA-A receptors of the same superfamily are CRELD1-independent PMID: 30407909. Conservation is demonstrated — mouse Creld1 rescues *C. elegans crld-1a* mutants, and Creld1 knockdown in mouse muscle cells decreases AChR surface expression.

Immune system involvement

"*We here provide evidence that CRELD1 is an important gatekeeper of immune system homeostasis*" PMID: 33169013; the accompanying commentary is titled *"CRELD1 controls T cell homeostasis"* PMID: 33169012. Low-CRELD1 individuals show reduced CD4 T cells. This underlies the recurrent-infection phenotype, though baseline PBMC subsets in the JELANS cohort were broadly similar to healthy donors — suggesting a subtle or functional (rather than numeric) immune defect.

Subcellular compartment

The **endoplasmic reticulum** (GO:0005783) is the central organelle: CRELD1 operates there both as an NFAT-signaling regulator and as a receptor-maturation chaperone.

Suggested cell types (CL)

Endocardial cell (CL:0002350); cardiomyocyte (CL:0000746); skeletal myocyte/myotube (CL:0000188); motor neuron (CL:0000100); neuron (CL:0000540); CD4+ T cell (CL:0000624).

Molecular profiling

A large human expression-variance cohort defined CRELD1-dependent transcriptional/immune phenotypes PMID: 33169013. No JELANS-specific human-tissue omics atlas yet exists.

Section 7 — Anatomical Structures Affected

Organ level. - *Primary:* brain/CNS (UBERON:0000955) — epilepsy, developmental delay, hypotonia; heart (UBERON:0000948) — arrhythmia, septal/structural defects; skeletal muscle (UBERON:0001134) and neuromuscular junction (UBERON:0002409) — hypotonia, weakness, arthrogryposis. - *Secondary/complications:* joints/limbs (arthrogryposis, UBERON:0000982); immune system (recurrent infections). - *Body systems:* nervous, cardiovascular, musculoskeletal/neuromuscular, immune.

Tissue and cell level. Nervous tissue (cholinergic circuits); cardiac endocardium (endocardial cells) and myocardium (cardiomyocytes); endocardial cushions/valves (UBERON:0003709); skeletal muscle; CD4+ T lymphocytes. See CL terms in Section 6.

Subcellular level. Endoplasmic reticulum (GO:0005783) is the key compartment; CRELD1 is a membrane-associated ER protein. Plasma-membrane receptor abundance (at the NMJ/synapse) is the downstream readout.

Localization / lateralization. Manifestations are **generalized/bilateral** (generalized hypotonia, generalized epilepsy, symmetric arthrogryposis); cardiac defects are central (septal). No consistent lateralization is described.

Section 8 — Temporal Development

Onset. Congenital to early infantile. Hypotonia and structural cardiac defects are congenital; epilepsy is typically **neonatal to first year of life** and frequently intractable PMID: 37947183; PMID: 40980404. Arthrogryposis (when present) is congenital.

Progression. The neurodevelopmental course shows **developmental plateauing** rather than frank regression, while **non-neurologic complications are slowly progressive** (cardiac rhythm disturbances, infections). The defining cohort describes these complications "in survivors," indicating the disorder can be **life-limiting/fatal in a subset**. The course is therefore best described as developmental with slowly progressive multisystem complications, with variable severity.

Patterns. Seizures may be episodic and, in milder cases, amenable to control; arrhythmias are episodic. No spontaneous remission of the underlying disorder. **Critical periods:** the prenatal/perinatal and first-year windows (cardiac morphogenesis; NMJ/synapse formation; seizure onset) are the key vulnerability windows and plausible intervention windows.

Section 9 — Inheritance and Population

Inheritance. Autosomal **recessive** (biallelic *CRELD1*). Parents are unaffected carriers.

Penetrance / expressivity. Appears **highly penetrant** for a neurodevelopmental phenotype in biallelic individuals, with **variable expressivity** — ranging from predominantly neurodevelopmental to severe neuromuscular/arthrogrypotic presentations (cardiac arrhythmia ~50%, infections in a subset).

Genetic anticipation / mosaicism. Not applicable (not a repeat-expansion disorder); no germline mosaicism reports.

Founder effects / consanguinity. The recurrent **p.(Cys192Tyr)** allele in 10/14 families suggests a **founder-like effect**; consanguinity may contribute to biallelic (including homozygous) genotypes.

Epidemiology. JELANS is **ultra-rare**. Only ~18 individuals from 14 families (defining cohort) plus isolated subsequent cases are reported; true prevalence/incidence are unknown and presumably <1 per 1,000,000 (not available).

Population demographics. The defining cohort was international; no specific ethnic predilection is established beyond the founder-like recurrent allele. **Sex ratio** is not clearly skewed (consistent with autosomal recessive inheritance). Age distribution reflects a pediatric-onset disorder. **Carrier frequency** is not formally established; individual pathogenic alleles are rare in gnomAD.

Section 10 — Diagnostics

Genetic testing is the definitive diagnostic modality. - *Recommended approach:* **trio exome (WES) or genome (WGS) sequencing**, which identifies biallelic *CRELD1* variants and establishes phase. The syndrome was defined by next-generation sequencing plus GeneMatcher-style international matchmaking PMID: 37947183. - *Single-gene / trio testing:* confirm biallelic *CRELD1* variants and the **trans** configuration (typically frameshift + missense); confirm/segregate the recurrent p.(Cys192Tyr) and family-specific frameshift alleles. - *Gene panels:* epilepsy, neurodevelopmental, hypotonia, arthrogryposis, and congenital-myasthenic-syndrome panels should include *CRELD1*. - *CMA/karyotype/FISH:* not diagnostic (single-gene disorder) but used to exclude chromosomal differentials. mtDNA and repeat-expansion testing are not indicated.

Clinical / ancillary tests. - *EEG* for epilepsy characterization (often shows refractory epileptiform activity). - *ECG / Holter / echocardiography* for arrhythmia and structural cardiac screening (arrhythmia ~50%, structural defect ~10%). - *Brain MRI* for developmental delay/epilepsy evaluation (findings not yet systematically characterized). - *Neuromuscular work-up* (EMG/repetitive nerve stimulation, CK) in the CMS-like/arthrogryposis presentation. - *Immune evaluation* (immunoglobulins, lymphocyte subsets) given recurrent infections — note baseline PBMC subsets were grossly normal in the cohort.

Biomarkers. No specific circulating/biochemical biomarker; diagnosis rests on genotype plus phenotype. No methylation episignature defined.

Clinical criteria / differential diagnosis. No formal consensus criteria yet. Differential diagnoses include other recessive developmental-and-epileptic encephalopathies; **congenital myasthenic syndromes** (nAChR-subunit, agrin/rapsyn/DOK7-related) — especially for the weakness/arthrogryposis presentation; *NFATC1*-related AVSD (distinct, cardiac-restricted) PMID: 30007050; and syndromic causes of arthrogryposis. Distinguished by biallelic *CRELD1* variants.

Screening. No population newborn screening. **Cascade carrier testing** of relatives and **prenatal/preimplantation genetic testing** are appropriate once familial variants are known.

Section 11 — Outcome / Prognosis

Survival/mortality. No formal survival statistics, but the defining cohort explicitly refers to "survivors," indicating **the disorder is life-limiting in a subset** PMID: 37947183. Principal risks are early-onset/refractory epilepsy, severe hypotonia/respiratory compromise, and cardiac arrhythmia (potentially life-threatening if unmonitored).

Morbidity / function. High morbidity: lifelong intellectual disability, refractory epilepsy, hypotonia/weakness (up to arthrogryposis with profound weakness), plus the need for ongoing cardiac and infection surveillance. Significant disability and care-dependence are expected in moderate-to-severe cases.

Disease course / complications. Developmental plateauing; slowly progressive cardiac rhythm disturbances; recurrent infections; refractory seizures. Structural heart defects (~10%) may require surgical repair.

Prognostic factors. Severity and control of epilepsy, presence/severity of cardiac arrhythmia, the severe neuromuscular phenotype (arthrogryposis + refractory epilepsy portends a worse course), and the **residual function of the missense allele** (genotype). No validated molecular prognostic biomarker exists.

Section 12 — Treatment

Current standard: supportive / multidisciplinary. There is no disease-modifying or curative therapy. Management targets each system: - *Epilepsy:* antiseizure medications chosen by seizure type/EEG (NCIT: Anticonvulsant Agent); seizures are frequently **intractable/therapy-refractory** PMID: 40980404. - *Cardiac:* antiarrhythmic/cardiology-directed therapy; device therapy (pacemaker) if indicated; surgical repair of structural defects. - *Neuromuscular/developmental:* physical, occupational, and speech therapy (NCIT: Physical Therapy; Occupational Therapy; Speech Therapy); management of hypotonia and contractures; feeding support. - *Immune/infectious:* prompt treatment and surveillance of recurrent infections; immunology referral; consider immunoglobulin replacement if a humoral defect is found (individualized).

Mechanism-based (candidate, unproven) therapy — cholinesterase inhibitors. Because CRELD1 is a **maturation enhancer controlling surface abundance of nicotinic AChRs** PMID: 30407909, biallelic loss reduces functional AChRs at the NMJ, producing a **congenital-**

myasthenic-syndrome-like picture. This mechanistically motivates **cholinesterase inhibitors such as pyridostigmine** (NCIT: Pyridostigmine; a reversible AChE inhibitor that raises synaptic acetylcholine), by analogy to AChR-deficiency CMS PMID: 40980404. **Important caveat:** this is an untested mechanistic inference requiring genetic-subtype caution — pyridostigmine is beneficial in some CMS subtypes but harmful in others (e.g., slow-channel, DOK7, COLQ-related). Any trial should be cautious, monitored, and ideally protocolized.

Advanced / targeted / gene therapy. None available; no gene, cell, RNA, or targeted small-molecule therapy for *CRELD1* (research stage only; not applicable clinically). No registered clinical trials specific to JELANS were identified.

Treatment strategy. Multidisciplinary care (neurology, cardiology, immunology, developmental pediatrics, genetics). The AChR-deficiency mechanism provides the main precision-medicine hypothesis (cholinesterase inhibitors, with subtype caution); no established pharmacogenomic protocol exists for this gene.

Section 13 — Prevention

- **Primary prevention:** Not applicable for a Mendelian disorder beyond reproductive options. **Genetic counseling** for at-risk couples (25% recurrence risk per pregnancy for carrier couples).
- **Secondary prevention (reproductive):** Carrier screening, cascade testing, prenatal diagnosis, and **preimplantation genetic testing (PGT-M)** for families with known biallelic variants. Early sequencing-based diagnosis enables early seizure control, cardiac surveillance, and infection management.
- **Tertiary prevention (preventing complications):** Proactive cardiac surveillance (ECG/echo) for arrhythmia; structured seizure management; infection monitoring/immunology follow-up; physical therapy to limit contractures.
- **Counseling:** Autosomal recessive recurrence risk is **25%**; cascade testing and reproductive options should be offered.
- **Immunization / public health / environmental:** Standard childhood immunizations and infection precautions for the infection-prone subset; no disease-specific vaccine or environmental intervention applies (not applicable).

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs.** *CRELD1* is evolutionarily conserved. Orthologs: mouse *Creld1* (NCBI Gene 171508; NCBITaxon:10090), *Xenopus tropicalis* creld1 (NCBITaxon:8364), and the single *C. elegans* gene *crld-1* (NCBITaxon:6239), whose two isoforms correspond to the vertebrate paralogs CRELD1 (transmembrane) and CRELD2 (KDEL) PMID: 30407909.
- **Natural disease in other species.** No naturally occurring companion-animal or wildlife disease equivalent to JELANS is documented in OMIA (not available).
- **Comparative biology.** Mouse *Creld1* is essential for heart development; **global knockout is embryonic lethal** PMID: 24697899; conditional KO reveals myocardial roles PMID: 33773996. *Xenopus* creld1 knockdown reproduces developmental defects and seizure susceptibility PMID: 37947183. The AChR-maturation function is **deeply conserved**: mouse Creld1 rescues *C. elegans crld-1a* mutants, and mouse-muscle Creld1 knockdown lowers surface AChR PMID: 30407909. Both the calcineurin/NFATc1 and the ER-PDI/AChR mechanisms are conserved — supporting cross-species validity.
- **Transmission / zoonosis.** Not applicable (genetic disorder).

Section 15 — Model Organisms

Model	Type	Key findings	Recapitulation	Reference
C. elegans — *crld-1* mutants	Invertebrate	Identified in a forward (levamisole-resistance) screen; required for synaptic nicotinic AChR expression; crld-1A is an ER PDI assembling AChRs; rescued by mouse Creld1	Core neuromuscular/ AChR-assembly arm; proves conservation	PMID: 30407909
C. elegans — patient-variant knock-in	Invertebrate	Compound-heterozygous CRELD1 variants reduce AChRs at the NMJ	Models patient genotype's functional effect	PMID: 40980404
Xenopus tropicalis — creld1 knockdown/ CRISPR	Vertebrate	Developmental defects; increased induced-seizure susceptibility; ~10% structural cardiac defects; overexpression assay measures missense allele function	Strong neurodevelopmental/ seizure recapitulation; key disease-validation model	PMID: 37947183
Mouse — Creld1 knockdown in muscle cells	In vitro (mammalian)	Decreased AChR surface expression	Supports nAChR-maturation mechanism in mammals	PMID: 30407909
Mouse — global *Creld1* KO	Mammalian, knockout	Embryonic lethal; essential for heart development/ endocardial proliferation	Cardiac developmental role; lethality limits postnatal study	PMID: 24697899
Mouse — conditional	Mammalian, conditional		Enables tissue-specific cardiac study	PMID: 33773996

Model	Type	Key findings	Recapitulation	Reference
Creld1 KO (endocardial/ myocardial)		Regulates myocardial development and function		
Mouse — Ts65Dn × *Creld1* LoF cross	Mammalian, genetic interaction	Increases CHD frequency on trisomic background	Models CRELD1 as a CHD modifier (AVSD context)	PMID: 22523272

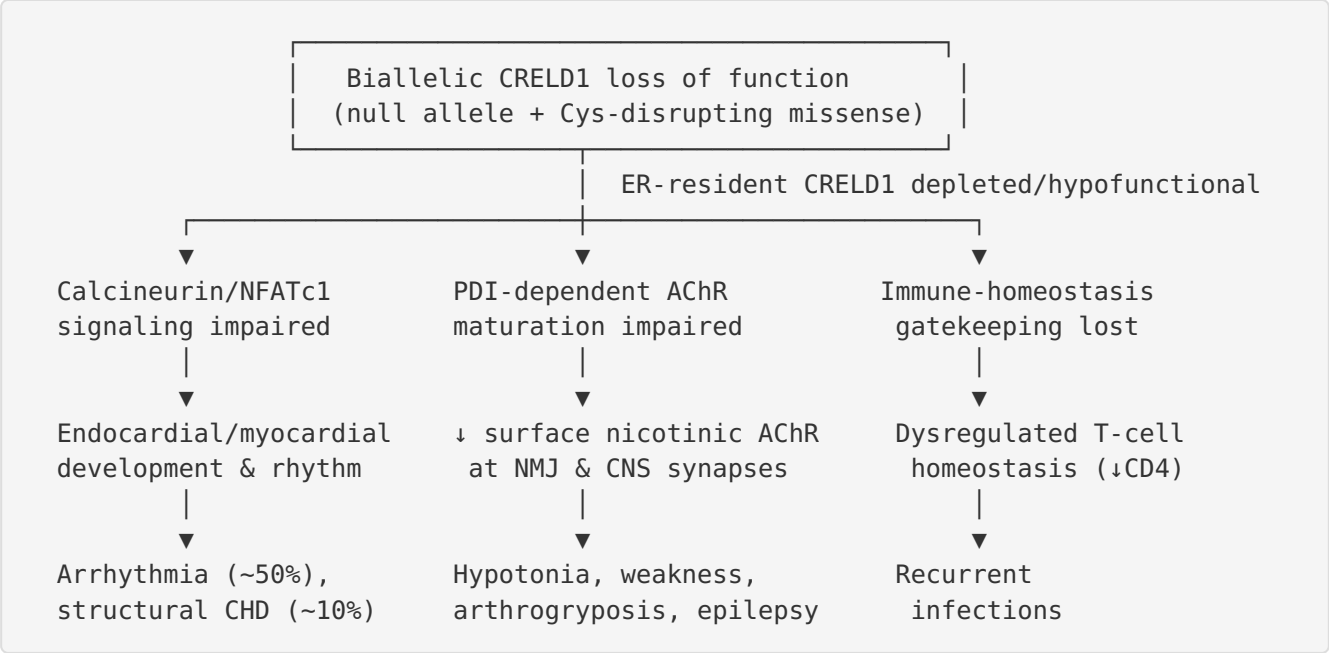
Model resources. MGI (mouse *Creld1*), Xenbase (Xenopus creld1), WormBase (*crld-1*), Alliance of Genome Resources.

Model strengths. Collectively, the models reproduce the cardiac (mouse NFAT), neuromuscular/AChR (worm, mouse muscle cells), and seizure/developmental (Xenopus) arms, and demonstrate cross-species rescue — strong evidence of conserved mechanism.

Model limitations. Global mouse KO is embryonic lethal, precluding study of postnatal neurodevelopment and epilepsy; no mouse model fully recapitulates the combined human neuro-cardiac-immune triad with the specific biallelic null/hypomorphic genotype. Patient-specific iPSC/organoid or humanized knock-in CNS models are not yet established.

Mechanistic Model / Interpretation

JELANS is best understood as a **dual-mechanism, ER-centered CRELD1 loss-of-function disorder**:



The cardiac arm (NFAT) and the neuromuscular/CNS arm (AChR maturation) are both experimentally grounded; the immune arm is supported by independent mouse/human work. The convergence on the **endoplasmic reticulum** — where CRELD1 both scaffolds calcineurin/NFAT signaling and performs PDI-dependent receptor assembly — unifies the otherwise disparate neuro-cardiac-neuromuscular-immune phenotype. The recurrent loss of **cysteine residues** (Cys192, Cys292) in pathogenic missense alleles is mechanistically coherent with disruption of CRELD1's disulfide-isomerase chemistry, which both functions depend on.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
37947183	*Biallelic CRELD1 variants cause a multisystem syndrome...*	**Defining cohort** (18 pts/14 families); inheritance, genotype, phenotype frequencies, Xenopus + functional validation, natural history	Human clinical + model
40980404	*Biallelic CRELD1 variants cause severe muscle weakness and infantile epilepsy*	Phenotype extension (arthrogryposis, refractory epilepsy); p.Cys292Tyr + frameshift; C. elegans NMJ AChR reduction; CMS-like therapeutic rationale	Human clinical + C. elegans
30407909	*CRELD1 is an evolutionarily-conserved maturational enhancer of ionotropic AChRs*	**Core neuromuscular mechanism**: ER PDI, AChR maturation, cross-species conservation, client specificity	C. elegans / mouse / in vitro
24697899	*Murine Creld1 controls cardiac development through calcineurin/ NFATc1*	**Cardiac mechanism**: NFAT signaling; KO embryonic lethal	Mouse / in vitro
33773996	*Creld1 regulates myocardial development and function*	Conditional KO; tissue-specific cardiac role	Mouse
33169013	*CRELD1 modulates homeostasis of the immune system*	Immune-gatekeeper function → infections; ↓CD4 T cells	Human cohort / mouse
33169012	*CRELD1 controls T cell homeostasis* (commentary)	Supports T-cell homeostasis role	Review
22523272	*Genetic modifiers in Down syndrome CHD*	Context: CRELD1 as dosage-threshold CHD modifier	Mouse / human
30007050	*NFATC1 missense mutations and AVSD*	Supports calcineurin/NFAT pathway in cardiac defects; differential	Human / zebrafish
36929416			

PMID	Title (abbrev.)	Role	Evidence type
	CRELD1 variants and bicuspid aortic valve in Turner syndrome	Context: heterozygous modifier role (distinct from JELANS)	Human genetics
29054759; 22987595; 21413875; 24927998	Heterozygous CRELD1 variants as AVSD risk/ modifier alleles	Context: contrasting heterozygous/ GoF modifier biology	Human genetics / in vitro

Evidence spans **human clinical** (defining cohort, case reports), **model organism** (mouse, Xenopus, C. elegans), and **in vitro** (muscle-cell knockdown, functional missense assays). The mechanistic model is unusually well supported for an ultra-rare disorder because CRELD1's two core functions were independently established before JELANS was delineated.

Supported vs. Refuted Hypotheses

- **Supported:** JELANS is autosomal recessive, caused by biallelic (null + hypomorphic cysteine-missense) *CRELD1* variants; it is a multisystem neuro-cardiac-neuromuscular-immune syndrome; mechanisms involve impaired calcineurin/NFATc1 signaling **and** impaired ER maturation of nicotinic AChRs; the p.Cys192Tyr allele is recurrent (10/14 families); model organisms (C. elegans, Xenopus, mouse) recapitulate AChR loss, seizure susceptibility, and cardiac defects.
- **Newly strengthened:** The AChR-maturation mechanism is now **demonstrated** (not merely inferred), providing a unified explanation for the neuromuscular and epilepsy features and a **mechanism-based (candidate) therapeutic rationale for cholinesterase inhibitors** (pyridostigmine), with subtype caution.
- **Nuanced / partially refuted:** Recurrent infections are *not* explained by a gross baseline shift in patient PBMC immune-cell proportions (similar to healthy donors); the immune defect is likely functional rather than numeric. The course is **not purely static** — it shows developmental plateauing with slowly progressive complications and is **life-limiting in a subset**.

Limitations and Knowledge Gaps

1. **Small cohort.** The clinical picture rests on ~18 individuals plus isolated cases; frequencies (~50% arrhythmia, ~10% structural CHD) are approximate and may shift as more patients are identified. Penetrance, prognosis, sex ratio, and natural history are preliminary.
2. **No human disease-tissue omics.** No transcriptomic/proteomic/metabolomic/epigenomic atlas of patient tissues exists; molecular profiling evidence is indirect (model systems, immune-homeostasis cohort).
3. **Epilepsy and immune mechanisms partly inferred.** The causal link from AChR deficiency to *central* refractory epilepsy and the NFAT→neurodevelopment link are inferred rather than directly demonstrated in human CNS; the immune defect appears functional rather than numeric.
4. **No validated biomarker or prognostic model.** Genotype-phenotype correlations (missense residual function; severe neuromuscular subtype) are suggestive but unquantified; no methylation episignature.
5. **Therapy untested.** The cholinesterase-inhibitor rationale is a mechanistic hypothesis; efficacy and safety in JELANS are unproven, and CMS experience warns that AChE inhibitors harm some subtypes.
6. **No faithful mammalian CNS model.** Embryonic lethality of global mouse KO limits study of the neurodevelopmental/epileptic phenotype; conditional neural and patient-genotype knock-in CNS models are lacking.

Proposed Follow-up Experiments / Actions

1. **Expand and deep-phenotype the cohort** via international registries and GeneMatcher to firm up phenotype frequencies, natural history, survival, sex ratio, and genotype-phenotype correlation (especially missense residual-function vs severity).
2. **Build a patient-genotype knock-in mouse or neural conditional KO** to model epilepsy and neurodevelopment; complement with **patient iPSC-derived motor neurons/muscle and cerebral/neuromuscular organoids** to measure surface AChR abundance and synaptic function.
3. **Quantify AChR deficits in patient-derived systems** and test whether **cholinesterase inhibitors (pyridostigmine) or 3,4-diaminopyridine** rescue neuromuscular function —

and whether any subtype would be harmed — before any cautious, protocolized human trial.

4. **Electrophysiology in patients** (repetitive nerve stimulation, single-fiber EMG) to test for a decremental/NMJ-transmission defect supporting the CMS-like model and guiding AChE-inhibitor trials.
5. **Define a methylation episignature** and search for diagnostic/prognostic biomarkers to aid VUS reclassification; estimate gnomAD-based carrier frequency.
6. **Structural/biochemical work** on CRELD1 PDI activity with cysteine-disrupting missense alleles (Cys192Tyr, Cys292Tyr) to confirm the disulfide-chemistry mechanism and prioritize alleles amenable to chaperone-rescue strategies.
7. **Detailed immune characterization** (T-cell subsets/function) in patients to clarify the infection phenotype and whether prophylaxis is warranted.

Evidence source types are indicated throughout: human clinical (defining cohort and case reports), model organism (mouse, Xenopus, C. elegans), and in vitro (muscle-cell knockdown, functional missense assays). Causal steps that are inferred rather than directly demonstrated are explicitly flagged.
